Monotonic Array

AlgoExpert · Medium · Arrays

The problem

- Is the array entirely non-increasing or entirely non-decreasing?

Intuition

- An array is monotonic if it never goes both up and down.

Brute force vs optimal

- Naive: determine direction from the first unequal pair, then re-scan
- Simpler: track two flags in one pass → O(n)
- No sorting or extra memory needed

Approach

- 1. Track two flags, non-increasing and non-decreasing
- 2. Clear the opposite flag on each rise or fall
- 3. Return true if either flag survives

Worked example




```
-1 → -5
non-decreasing possible: False
non-increasing possible: True
```

-1 → -5: inc=False, dec=True

Worked example




```
-10 → -1100
non-decreasing possible: False
non-increasing possible: True
```

-10 → -1100: inc=False, dec=True

Worked example




```
-1100 → -1100
non-decreasing possible: False
non-increasing possible: True
```

-1100 → -1100: inc=False, dec=True

Worked example




```
-1102 → -9001
non-decreasing possible: False
non-increasing possible: True
```

-1102 → -9001: inc=False, dec=True

Java solution

```java
public class MonotonicArray {
    public static boolean isMonotonic(int[] a) {
        boolean inc = true, dec = true;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[i - 1]) inc = false;
            if (a[i] > a[i - 1]) dec = false;
        }
        return inc || dec;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Empty or 1-element arrays are monotonic
- Equal neighbours are allowed
- Leading equal values before the trend

Interview tips & follow-ups

- Early exit when both flags are false
- Follow-up: strictly monotonic

Recap

- An array is monotonic if it never goes both up and down.
- Complexity: O(n)